

Recently graduated MSc in Molecular Biomedicine (University of Copenhagen, thesis 12/12) with a background in **drug delivery and pharmaceutical biophysics**, focused on **oral peptide delivery, lipid/polymeric nanocarriers, epithelial permeation, and molecular dynamics of self-assembling systems**. My master's thesis combined experimental release and Ussing-chamber permeability studies with atomistic and coarse-grained (MARTINI) simulations of nanocarriers on HPC. Comfortable with biophysical characterisation (DLS, zeta potential, ATR-FTIR, NMR), scientific programming (Python, R, MATLAB), and HPC/Linux workflows, and motivated to apply biophysical and computational methods to the molecular mechanisms of permeation enhancers and oral peptide delivery.

Education

2023 – 2025 MSc Molecular Biomedicine, Copenhagen University, Denmark, **GPA (11/12)**

courses: Molecular pathology, Statistics for biomedicine, Data science for genomics (R, Sequencing analysis, PCA, statistical tests), Medicinal and Biostructural chemistry, CRISPR design and hands-on gene editing

2019 – 2023 BSc Biotechnology, Yazd University, Yazd, Iran, **GPA 18.31/20 (3.76/4)**

Relevant courses: Biochemistry, biophysical chemistry, medicinal biotechnology, biomaterial, nanobiotechnology

Research interests

- Membrane biophysics & drug delivery
- Oral peptide delivery & permeation
- Molecular dynamics (atomistic & CG)
- Biophysical characterisation
- Release kinetics & redistribution
- Scientific programming & data analysis

Relevant skills

Computational & Modelling Skills

- Molecular dynamics simulations: atomistic and coarse-grained (MARTINI) using GROMACS and LAMMPS, including HPC/cluster workflows on Linux
- Coarse-grained mapping of polymers with PolyPy (applied to hyaluronic acid-based nanogels)
- Computational chemistry: molecular docking, protein structure refinement, in-silico binding studies
- Protein structure prediction and modelling: AlphaFold/ColabFold, Rosetta, PyMOL, COOT (applied in peer-reviewed structural and X-ray refinement work)
- Scientific programming: Python, R, MATLAB (data analysis, statistical testing, PCA, curve fitting of release profiles)
- Release kinetics analysis and mathematical fitting of drug-release data (experimental and model-based)
- General: Linux terminal, GraphPad Prism, ImageJ, Microsoft Office

Drug Delivery, Biophysics & Formulation

- Epithelial permeation and release studies: Ussing chamber, pH-dependent release, permeability assays, Labbot
- Nanocarrier characterisation: DLS, zeta potential, NTA, ATR-FTIR, NMR spectroscopy, osmolarity
- Nanoparticle synthesis: polymeric nanogels (OSA-HA), lipid, magnetic and silica nanoparticles
- Microfluidic-based nanoparticle fabrication
- Hydrophobic cargo loading and quantification (Coumarin 6, doxorubicin) and peptide encapsulation
- Formulation strategies: lyophilization, coformulation, electrospraying

Wet-lab & Analytical Techniques

- Cell culture, MTT assay, real-time PCR, gel electrophoresis
- Optical and fluorescence microscopy
- X-ray crystallographic data analysis and structure refinement (COOT, PyMOL)
- Biological and tissue sample handling

Communication & Organisation

- Scientific writing for publications, reports, and grant proposals
- Reagent ordering, inventory management, and daily lab upkeep
- Teaching techniques and methodologies to new researchers (KU Copenhagen, 2024)
- Hosted workshops and webinars on molecular docking (Yazd University and Biotech Cell Laboratory, 2022)

Research experience

- **Master thesis** (September 2024 – August 2025) (Grade: 12/12):

Drug Delivery and Biophysics of Biopharmaceuticals group, Copenhagen University, Denmark

Environmental Behaviour and Release Kinetics of OSA-HA Nanogels for Oral Delivery: An Experimental and Computational Study

Developed pH-responsive OSA-modified hyaluronic acid nanogels for oral peptide delivery, combining microfluidic synthesis with experimental release/permeability studies and computational modelling. Ran epithelial permeability experiments in Ussing chambers, characterised environmental behaviour (DLS, zeta potential, ATR-FTIR, osmolarity) at gastric and intestinal pH, and investigated hydrophobic cargo loading (Coumarin 6, doxorubicin) alongside peptide release.

Skills: polymeric nanoparticle preparation, microfluidics, Ussing-chamber permeability, DLS / zeta potential, release kinetics analysis and curve fitting, lyophilization, coformulation, ATR-FTIR, hydrophobic cargo loading

- **Thesis preparation project** (July 2024 – September 2024) (Grade: 12/12):

Drug Delivery and Biophysics of Biopharmaceuticals group, Copenhagen University, Denmark

Development of Coarse-Grained Simulation Models to Predict Structural Dynamics of OSA Modified Hyaluronic Acid Nanogels

Built MARTINI coarse-grained models of OSA-modified hyaluronic acid using Polyply and GROMACS to predict nanogel self-assembly and structural dynamics, complementing atomistic MD and NMR-based structural validation. Work performed on HPC clusters under Linux.

Skills: MARTINI coarse-grained simulation, Polyply mapping, GROMACS, atomistic MD, NMR spectroscopy, HPC/Linux workflows, chemical modification of hyaluronic acid

- **Project outside course scope** (April 2024 – June 2024) (Grade: 10/12):

Receptor Structural Biology group (Kastrup group), Drug Design and Pharmacology Department, Copenhagen University, Denmark

Refinement of X-ray structure of the ligand-binding domain of the GluA2 flip variant (GluA2i-LBD) in the presence of glutamate and a positive allosteric modulator

Skills: X-ray diffraction data analysis, structure refinement, structural analysis, COOT, PyMOL

- **Internship at Biotech cell laboratory** (August 2022 – July 2024):

Isfahan, Iran

Comprehensive Repurposing of FDA-Approved Drugs to Target HPV E6 Protein: Integrating Docking Studies and Cellular Assays for Inhibitory Effect Evaluation

Skills: Molecular docking, docking structure analysis, cell culture, real-time PCR, MTT assay

- **Industrial Internship** (July 2021 – January 2022):

Behyar-sanat-sepahan corporation, Isfahan, Iran

Designing and fabrication of plasmid extraction kit using magnetic nanoparticles

Skills: Magnetic nanoparticle synthesis, silica nanoparticle synthesis, surface modification, electrophoresis gel, plasmid extraction

Coaching Experience

2024 Teaching techniques and methodologies to new researchers in the group, KU, Copenhagen, Denmark

2022 Holding a webinar on “Drug Design and Molecular Docking”, Yazd University, Iran

2022 Holding a workshop on “Drug Design and Molecular Docking”, Biotech Cell Laboratory, Isfahan, Iran

Languages

English (IELTS Score: 7 (C1)), Danish (Elementary level), Persian (Native)

Publications

Peer-Reviewed Papers

- 2024 Tabrizi F, Khatami M, Heidari MM, Bragança J, Tatari H, **Namnabat M**, Hadadzadeh M, Navabi Shirazi MA. "Novel and deleterious nucleotide variations in the HAND1 gene probably affect miRNA target sites and protein function in pediatric patients with congenital heart disease." *Molecular Biology Reports*.
- 2023 Khatami M, Ghazinader D, Ahmadi F, Heidari MM, Hadadzadeh M, **Namnabat M**. "Association of Pathogenic Missense Mutations of NKX2.6 gene in patients with Congenital Heart Disease: Bioinformatics Prediction Tools and Structural Interactions." *Gene Reports*.
- 2023 Chamani R, **Namnabat M**, Taleghani M, Imanpour A. "The RGD sequence improves the binding of collagen IV-derived peptides to integrins: an in-silico study." *Cellular and Molecular Research (Iranian Journal of Biology)*.
- 2022 Heidari MM, Khatami M, Kamalipour A, Kalantari M, Movahed M, Emmamy MH, Hadadzadeh M, Bragança J, **Namnabat M**, Mazrouei B. "Mitochondrial mutations in protein coding genes of respiratory chain including complexes IV, V, and mt-tRNA genes are associated risk factors for congenital heart disease." *Experimental and Clinical Science (EXCLI Journal)*.

Conference Papers

- 2022 **Namnabat M**, Chamani R. "Identification of human papilloma virus E6 oncoprotein inhibitor using docking simulation." 10th International Congress on Biology (IBC), Shahrekord, Iran.

Honors & Awards

- 2023 Member of the National Elite Organization of Iran.
- 2023 Graduated as a top student (rank 1 out of 40) in BSc Biotechnology, Yazd University.
- 2023 Accepted as an Elite Student in Tehran University for Master of Nanobiotechnology.
- 2022 Graduated BSc in the top 5% of the year.
- 2022 Ranked 4th in the National Olympiad of Stem Cell and Regenerative Tissue Engineering, Iran.

Certificates

- 2022 Drug Delivery System Design, International Institute of New Science, Iran
- 2020 Bioinformatics and Drug Design, Institute BIOCAMP, Iran
- 2020 Genome Editing Using CRISPR/Cas9, Yazd University, Iran
- 2020 SPSS Workshop, International Congress on Biomedicine, Iran
- 2020 CRISPR/Cas9 Workshop, International Congress on Biomedicine, Iran
- 2019 Primer Design and Real-Time PCR, Yazd University, Iran
- 2019 Workshop on Blood and Plant DNA Extraction, Yazd University, Iran
- 2019 Application of Nuclear Technology in Plant Modification and Biotechnology, Yazd University, Iran

References

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